

TBA4245 Geodesy

Example exercise no. 8. Multipath (MP)

It is important to be aware of the errors caused by multipath, M , as can give large errors on the GNSS measurements. Here the maximum error and the time of a period (cycle) of MP will be looked into.

See especially chapter 5.6 in the “GNSS” book by Hormann-Wellenhof’.

Measured signal intensity of the direct signal (S_D) and of the indirect signal (S_I) are shown in formula (5.151).

Measured signal intensity of the modulated signal (S_M) is shown in formula (5.152), and it can also be written as shown in formula (5.155). By comparing the two formulas, the formulas (5.158) and (5.159) between the direct signal and the modulated signal are obtained.

(a) **Multipath.** Maximum error caused by MP:

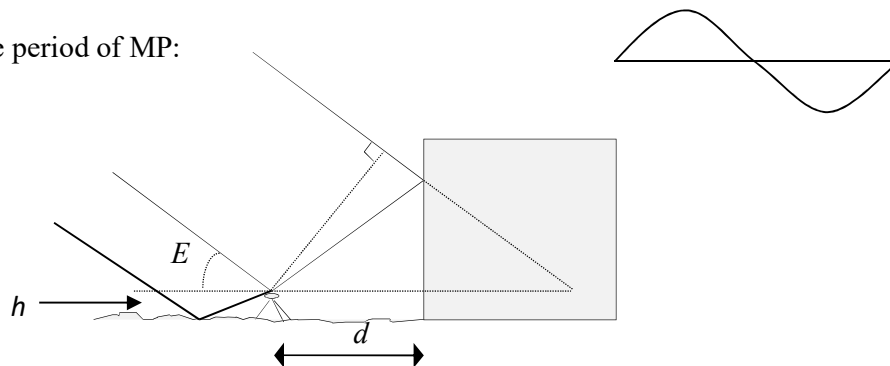
(a.1) Find the max value of the phase difference $\Delta\phi_M$.

We consider the damping factor β as a constant in this computation.

(a.2) Calculate the max value on the phase difference $\Delta\phi_M$ and on $\Delta\phi$. Which value of β must be used, and what is the corresponding value of β_M ? Interpret and evaluate the result!

(a.3) A period (cycle) is equal to 360° and is equal to 1 wavelength, λ . How long is $\Delta\phi_M$ in meters by phase measurements on the carrier waves?

(b) The period of MP:



E is the elevation of the incoming signals (height angle)

h is the height from terrain to the antenna, d is the distance to the reflecting plane.

The connections between MP, the elevation of the satellite, the distance to the reflecting surface and the wavelength of the signal are to be investigated. See the figure.

(b.1) Calculate how much longer (in meters) the indirect path is (Δs), both for d and h . (Use the figure above for d , and draw a similar figure for h).

(b.2) Transfer Δs to periods (cycles) of a wavelength, find the phase $\Delta\phi$, as 1 period in radians is equal to the phase 2π .

(b.3) Calculate the angular velocity of a GNSS satellite. Hint: Use the orbital time of the satellite.

(c) Numerical example:

The frequency of a wave is: $f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \cdot \frac{\partial \varphi}{\partial t}$, where ω is the angular velocity.

The elevation of the incoming signals is $E = 45^\circ$

Two values of d : $d_1 = 1$ meter and $d_2 = 10$ meters

For an average value of $\frac{\partial E}{\partial t}$ as seen from the observation site, use 50% of the angular speed of the satellite (taking into consideration the orbit of the satellite as seen from the observation site and how the satellite is moving relative to the receiver).

- (c.1) Deduct the formula of the actual frequency of the MP (you can consider the wavelength λ and d as constants).
- (c.2) Calculate the frequency of the MP for the two values of d .
- (c.3) Calculate the periods in minutes for MP for the two values of the distance d . Hint: Frequency is the number of period per second.
- (c.4) Evaluate the significance of the periodic behaviour of MP for the GNSS measurements. Tip: Measuring time in a point and the period of the multipath.

Solution:

(a) Multipath. Maximum error caused by MP:

- (a.1) Calculate the max value of the phase difference $\Delta\varphi_M$ and on $\Delta\varphi$.
We consider the damping factor β as a constant in this computation.

Differentiating formula (5.159):

$$\frac{\partial \Delta\varphi_M}{\partial \Delta\varphi} = 0 \quad \text{gives} \quad \Delta\varphi_{M,max} = \sin^{-1} \beta = \arcsin \beta = \text{inv sin } \beta$$

A little bit more explanation:

First finding $\Delta\varphi_{\max}$ from the differentiation above:

$$\cos \Delta\varphi = -\beta$$

Entering this value into formula (5.159) and using $\sin^2 + \cos^2 = 1$:

$$\tan \Delta\varphi_M = \frac{\sin \Delta\varphi_M}{\cos \Delta\varphi_M} = \frac{\beta}{(1 - \beta^2)^{0.5}}$$

which gives: $\sin \Delta\varphi_M = \beta$

(If $\sin \Delta\varphi_M = \beta$, then $\cos \Delta\varphi_M = (1 - \beta^2)^{0.5}$, see the formula on the line above, fits Ok)

(a.2) Calculate the max value on the phase difference $\Delta\varphi_M$ and on the $\Delta\varphi$. Which value of β must be used, and what is the corresponding value of β_M ? Interpret and evaluate the result!

Values of β and β_M :

$0 < \beta < 1$: The maximum value occurs when $\beta = 1$ (full reflex)

Formula (5.158), when $\cos \Delta\varphi = -\beta = -1$ (when max value, see (a.1)):

$$\beta_M = (1 + 1 - 2)^{0.5} = 0$$

Formula (5.161 and 5.162): $\Delta\varphi_M = \frac{1}{2} \cdot \Delta\varphi$

Thus, when: $\cos \Delta\varphi = -1$ (max value)

then: $\Delta\varphi = 180^\circ$

and: $\sin \Delta\varphi_M = 1$ (see (a.1)) gives $\Delta\varphi_M = 90^\circ$

or: $\Delta\varphi_M = \frac{1}{2} \cdot \Delta\varphi = 90^\circ$

Comments:

When $\beta = 1$ (full reflex) and the phase of the reflected signal is $\Delta\varphi = 180^\circ$ will, theoretically, the direct and the indirect signal cancel each other out, no signal is detected. Figure A below illustrates this (copied from the lecture notes). Figure B is illustrating that the modulated signal is weak when we are close to the max values of β and $\Delta\varphi$.

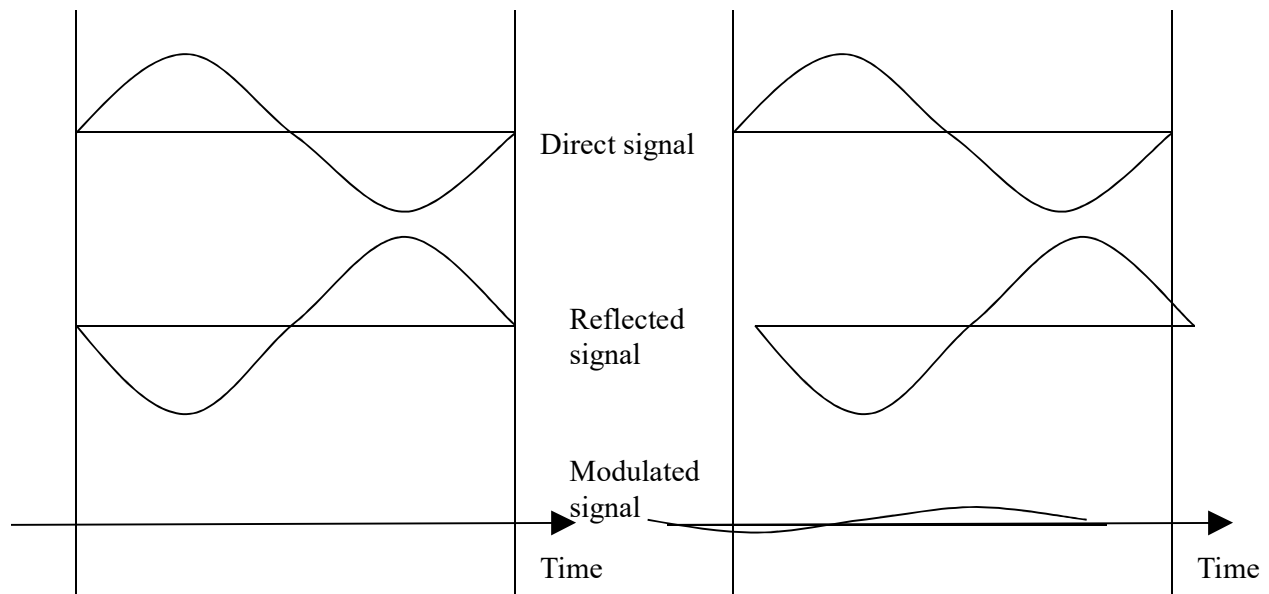


Figure A: $\Delta\varphi = 180^\circ$, $\beta = 1$
Gives $\Delta\varphi_M = 90^\circ$, $\beta_M = 0$

Figure B: $\Delta\varphi = 184^\circ$, β close to 1
Gives $\Delta\varphi_M = 92^\circ$, β_M close to 0

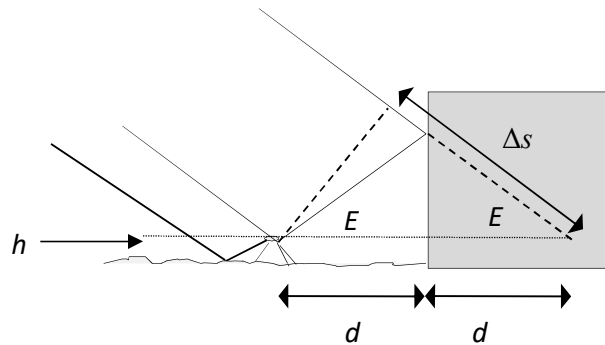
(a.3) A period (cycle) is equal to 360° and is equal to 1 wavelength, λ . How long is $\Delta\varphi_M$ in meters by phase measurements on the carrier waves?

The maximum error is when the modulated signal has a phase displacement (faseforskyvning) of $\Delta\varphi_M = 90^\circ$, which is equal to $\frac{1}{4}$ of the carrier wave λ .

For GPS with carrier waves of 19,2 cm on L1 and 24,0 cm on L2 in vacuum, the error of MP on the distance satellite-antenna will be: L1: 0,048 m, L2: 0,060 m.

(b) The period of MP:

(b.1) Calculate how much longer (in meters) the indirect path is (Δs), both for d and h . (Use the figure above for d , and draw a similar figure for h).



Reflecting plane above the antenna, the extra distance for the indirect signal:

From the figure: $\Delta s = 2d \cdot \cos E$

Reflecting plane below the antenna, the extra distance for the indirect signal:

From a similar figure (see figure 5.9 in the GNSS book):

$$\Delta s = 2h \cdot \sin E$$

(b.2) Transfer Δs to periods (cycles) of a wavelength, find the phase $\Delta\phi$, as 1 period in radians is equal to the phase 2π .

Δs as the number of periods (cycles) of the wave length: $\Delta s / \lambda$

Ratios, phases and distances: $\frac{\Delta\phi}{2\pi} = \frac{\Delta s}{\lambda}$

Solving the phase from the equation above: $\Delta\phi = 2\pi \cdot \Delta s / \lambda$

Solution, including substituting Δs from (b.1):

$$\text{Above the antenna, } d: \quad \Delta\phi = \frac{4\pi d}{\lambda} \cos E$$

$$\text{Below the antenna, } h: \quad \Delta\phi = \frac{4\pi h}{\lambda} \sin E$$

(b.3) Calculate the angular velocity of a GPS satellite. Hint: Use the orbital time of the satellite.

A GPS satellite is using ca. 12 h (more precise 11 h 58 min), one “round” is 2π radians.

$$\text{Angular velocity: } \omega = \frac{2\pi}{12 \cdot 60 \cdot 60} = 1,45 \cdot 10^{-4} \text{ rad / s}$$

(c) Numerical example:

The frequency of a wave is: $f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \cdot \frac{\partial \varphi}{\partial t}$, where ω is the angular velocity.

The elevation of the incoming signals is $E = 45^\circ$

Two values of d : $d_1 = 1$ meter and $d_2 = 10$ meters

For an average value of $\frac{\partial E}{\partial t}$ as seen from the observation site, use 50% of the angular speed of the satellite (taking into consideration the orbit of the satellite as seen from the observation site and how the satellite is moving relative to the receiver).

- (c.1) Deduct the formula of the actual frequency of the MP (you can consider the wavelength λ and d as constants).

Using the equations in (b.2) for $\Delta\varphi$:

$$\text{Above the antenna, } d: \quad \Delta\varphi = \frac{4\pi d}{\lambda} \cos E$$

$$\text{Below the antenna, } h: \quad \Delta\varphi = \frac{4\pi h}{\lambda} \sin E$$

We have to find an expression of the actual $\frac{\partial \varphi}{\partial t}$ and enter it into the frequency formula above

Above the antenna, d :

$$\frac{\partial(\Delta\varphi)}{\partial t} = -\frac{4\pi d}{\lambda} \cdot \sin E(t) \cdot \frac{\partial E(t)}{\partial t}, \text{ the absolute value is entered into the } f \text{ formula:}$$

The frequency of the multipath:

$$f = \frac{1}{2\pi} \cdot \frac{\partial \varphi}{\partial t} = \frac{2d}{\lambda} \cdot \sin E(t) \cdot \frac{\partial E(t)}{\partial t}$$

Below the antenna, h , similar deduction gives for the frequency of the multipath, see also formula (5.164) in the GNSS book:

$$f = \frac{\partial(\Delta\varphi)}{\partial t} = \frac{2h}{\lambda} \cdot \cos E(t) \cdot \frac{\partial E(t)}{\partial t}$$

The average value of $\frac{\partial E}{\partial t}$, 50% of the angular speed of the satellite, see (b.3):

$$\frac{\partial E(t)}{\partial t} = \frac{1}{2} \omega = 0,725 \cdot 10^{-4} \text{ rad/s}$$

- (c.2) Calculate the frequency of the MP for the two values of d .

$$\text{For L1 and } d = 10 \text{ m: } f = \frac{2d}{\lambda} \cdot \sin E(t) \cdot \frac{\partial E(t)}{\partial t} = \frac{2 \cdot 10}{0,192} \cdot \sin 45^\circ \cdot 0,725 \cdot 10^{-4} = 0,00534$$

$$\Rightarrow f = 5,34 \cdot 10^{-3} \text{ Hz}$$

See the values in the table below.

- (c.3) Calculate the periods in minutes for MP for the two values of the distance d . Hint: Frequency is the number of period per second.

For L1 and $d = 10$ m: The period is $p = 1 / f = 187 \text{ sec} \approx 3 \text{ min}$

See the values in the table below.

Frequency of MP and the period in minutes of the MP for the two values of the distance d :

	$d_1 = 1 \text{ m}$	$d_1 = 1 \text{ m}$	$d_2 = 10 \text{ m}$	$d_2 = 10 \text{ m}$
Carrier wave	Frequency, f	Period, p	Frequency, f	Period, p
L1	$5,4 \cdot 10^{-4} \text{ Hz}$	31 min	$5,34 \cdot 10^{-3} \text{ Hz}$	3 min
L2	$4,2 \cdot 10^{-4} \text{ Hz}$	40 min	$4,2 \cdot 10^{-3} \text{ Hz}$	4 min

- (c.4) *Evaluate the influence of the periodic behaviour of MP on the GNSS measurements.*
Tip: Measuring time in a point and the period of the multipath.

Max MP can be several cm in size (up to 6 cm, see (a.2)). We should carry out the measurements in a point for a time period that is minimum covering the whole cycle/period of MP.

Using real time measurements (RTK) with an observation time of < 10 seconds and even when using the method short-time-static with observation time of 5-10 minutes:

We might be measuring only at part of the period/cycle of the MP and thus have a systematic error on the observations.

This is especially critical when there is a surface reflecting most of the GNSS signal very close to the antenna.

See the textbook “GNSS” for how to eliminate the effect of multipath.